Enrollment No - 250843116002

Aim - Develop advanced visualizations (Box Plot, Heatmap, Pair Plot, Violin Plot).

Step 1 import requred libraries

```
#import pandas library
import pandas as pd
#import numpy library
import numpy as np
#import matlpotlib.pyplot library
import matplotlib.pyplot as plt
#import seaborn library
import seaborn as sns
```

```
#Create a dataset
data = {
    "Student_id": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14],
    "Study_Hours": [1.5,2.0,2.5,3.0,3.5,4.0,4.5,5.0,5.5,6.0,6.5,7.0,8.0,9.0],
    "Attendance": [65,70,72,75,78,80,82,85,87,90,91,93,95,97],
    "Midterm_Marks": [55,60,62,65,68,72,75,78,80,84,87,90,94,98],
    "Final_Marks": [45,50,52,57,62,68,72,76,81,85,89,93,97,100],
}
df = pd.DataFrame(data)
print(df)
```

```
    Student_id  Study_Hours  Attendance  Midterm_Marks  Final_Marks
0            1          1.5          65             55           45
1            2          2.0          70             60           50
2            3          2.5          72             62           52
3            4          3.0          75             65           57
4            5          3.5          78             68           62
5            6          4.0          80             72           68
6            7          4.5          82             75           72
7            8          5.0          85             78           76
8            9          5.5          87             80           81
9           10          6.0          90             84           85
10          11          6.5          91             87           89
11          12          7.0          93             90           93
12          13          8.0          95             94           97
13          14          9.0          97             98          100
```

```
#Display first five values from the dataset.
df.head()
```

	Student_id	Study_Hours	Attendance	Midterm_Marks	Final_Marks
0	1	1.5	65	55	45
1	2	2.0	70	60	50
2	3	2.5	72	62	52
3	4	3.0	75	65	57
4	5	3.5	78	68	62

Next steps: Generate code with df | New interactive sheet

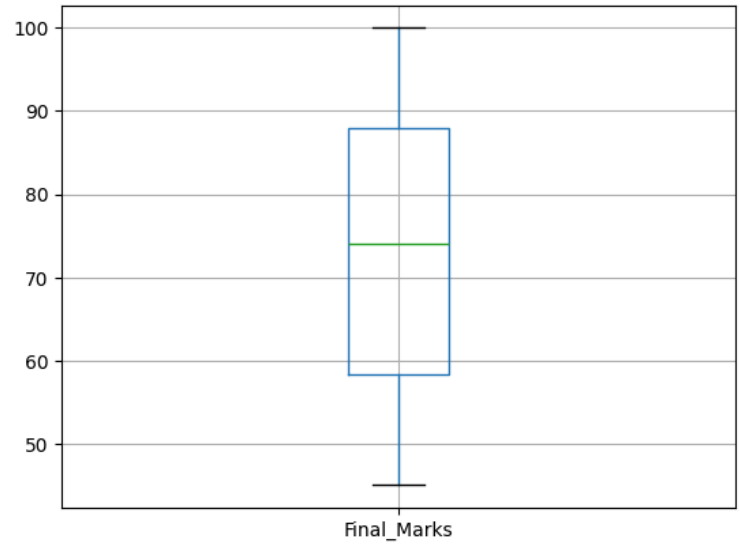
```
#Display the information about the detaset
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14 entries, 0 to 13
Data columns (total 5 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Student_id     14 non-null     int64
 1   Study_Hours    14 non-null     float64
 2   Attendance     14 non-null     int64
 3   Midterm_Marks  14 non-null     int64
 4   Final_Marks    14 non-null     int64
dtypes: float64(1), int64(4)
memory usage: 692.0 bytes
```

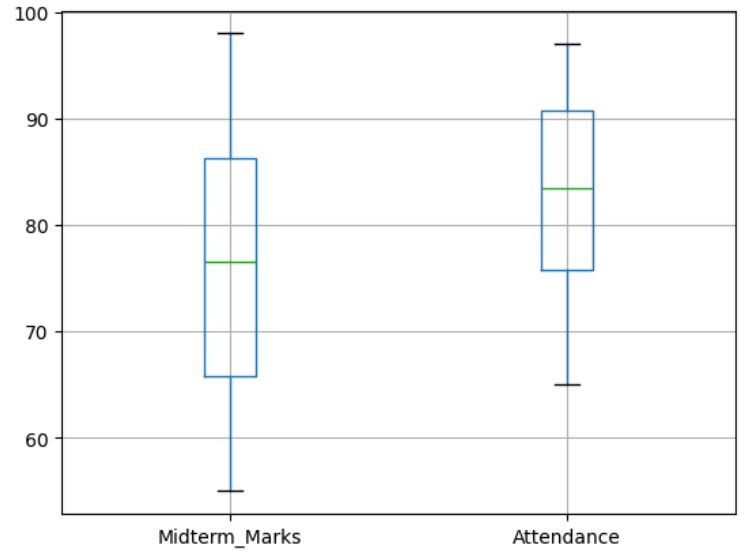
```
# Display the statistical information.
df.describe()
```

	Student_id	Study_Hours	Attendance	Midterm_Marks	Final_Marks
count	14.0000	14.000000	14.000000	14.000000	14.000000
mean	7.5000	4.857143	82.857143	76.285714	73.357143
std	4.1833	2.273836	9.944903	13.309503	18.257168
min	1.0000	1.500000	65.000000	55.000000	45.000000
25%	4.2500	3.125000	75.750000	65.750000	58.250000
50%	7.5000	4.750000	83.500000	76.500000	74.000000
75%	10.7500	6.375000	90.750000	86.250000	88.000000
max	14.0000	9.000000	97.000000	98.000000	100.000000

```
#Create a boox plot for final marks column.
df.boxplot(column="Final_Marks")
plt.show()
```




```
#Create box plot for multiple columns.
df.boxplot(column=["Midterm_Marks", "Attendance"])
plt.show()
```



Stap 4 - Create a Heatmap

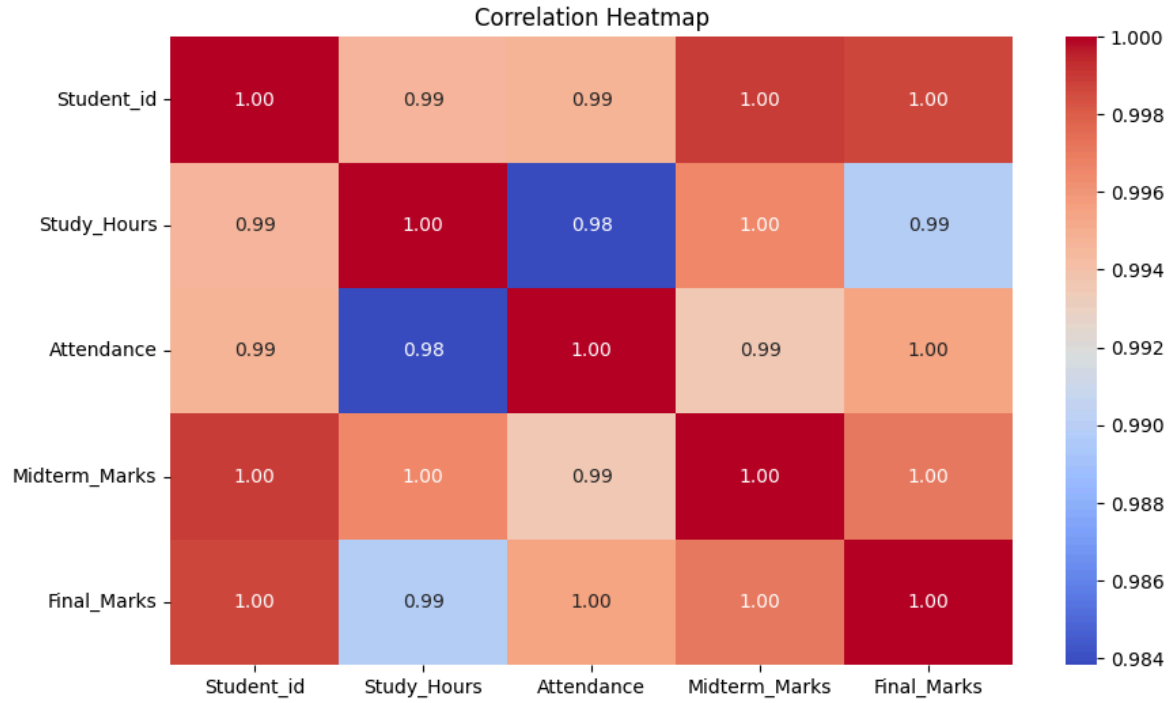
```
#select numerical columns for correction analysis from the dataset.
numerical_columns = df.select_dtypes(include=['number'])
```

```
correction_matrix = numerical_columns.corr()
print(correction_matrix)
```

```
              Student_id  Study_Hours  Attendance  Midterm_Marks  Final_Marks
Student_id      1.000000     0.994681    0.994762       0.998882     0.998612
Study_Hours     0.994681     1.000000    0.983822       0.996554     0.989873
Attendance      0.994762     0.983822    1.000000       0.993531     0.995492
Midterm_Marks   0.998882     0.996554    0.993531       1.000000     0.997041
Final_Marks     0.998612     0.989873    0.995492       0.997041     1.000000
```

```
# Calculate the correlation matrix.
correlation_matrix = numerical_columns.corr()

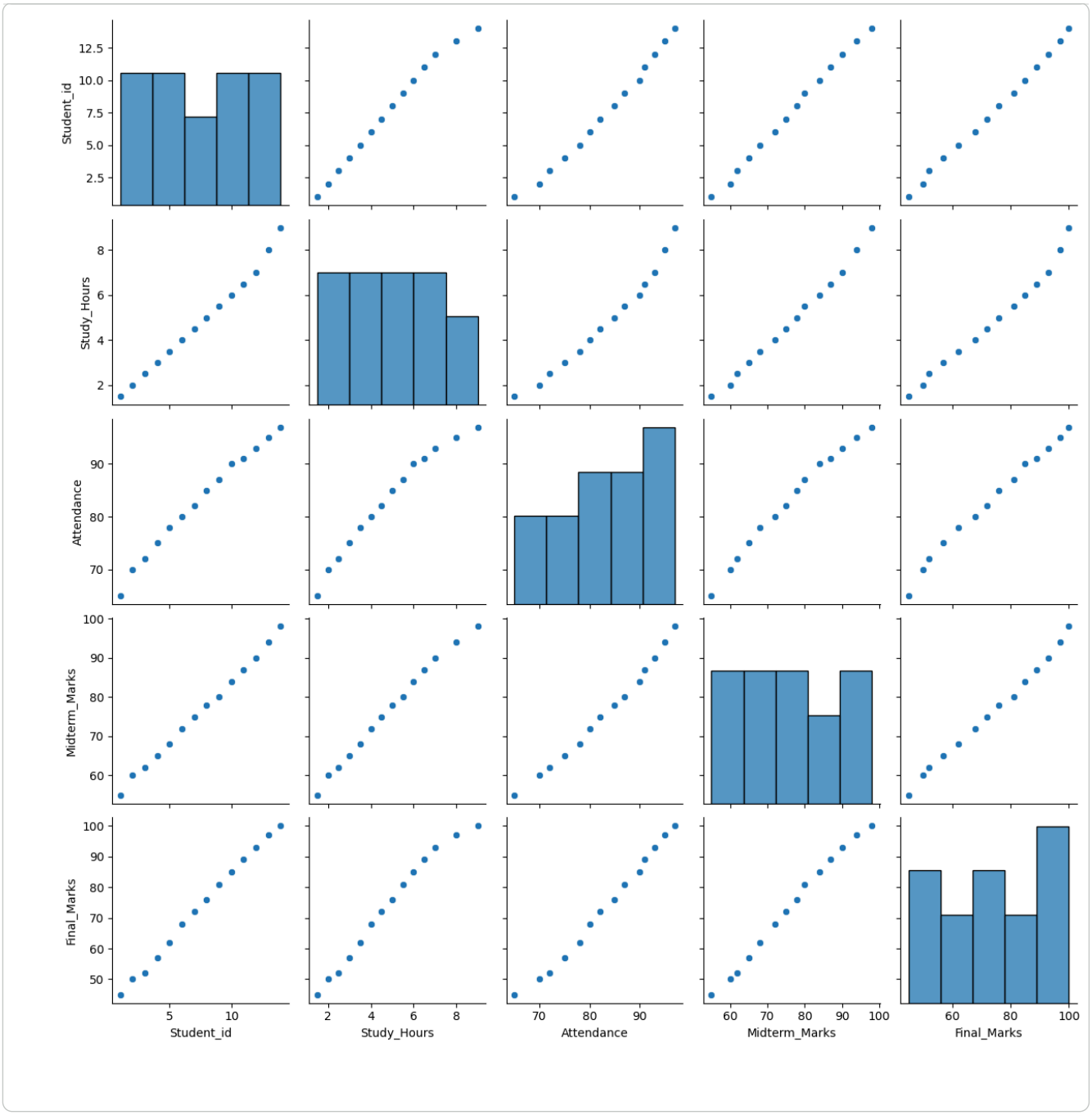
# Create the heatmap
plt.figure(figsize=(10, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Heatmap')
plt.show()
```



step - 5 Create a pair plot

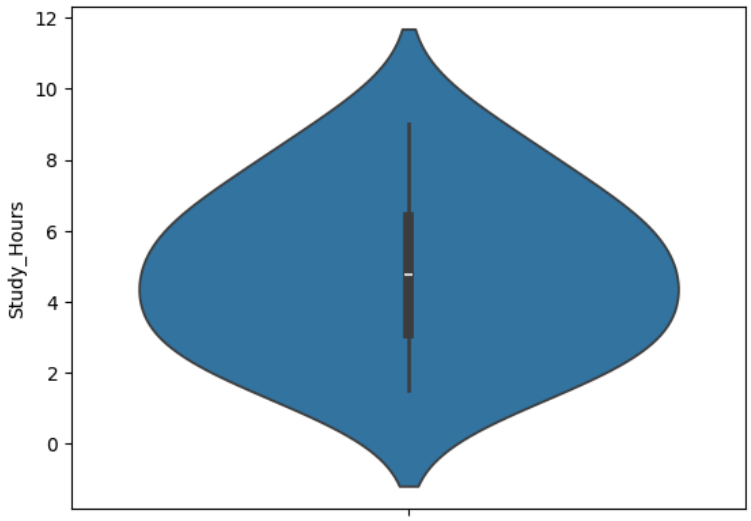
```
#select the numerical columns that we have to compare
numerical_columns = df.select_dtypes(include=['number'])

sns.pairplot(numerical_columns)
plt.show()
```

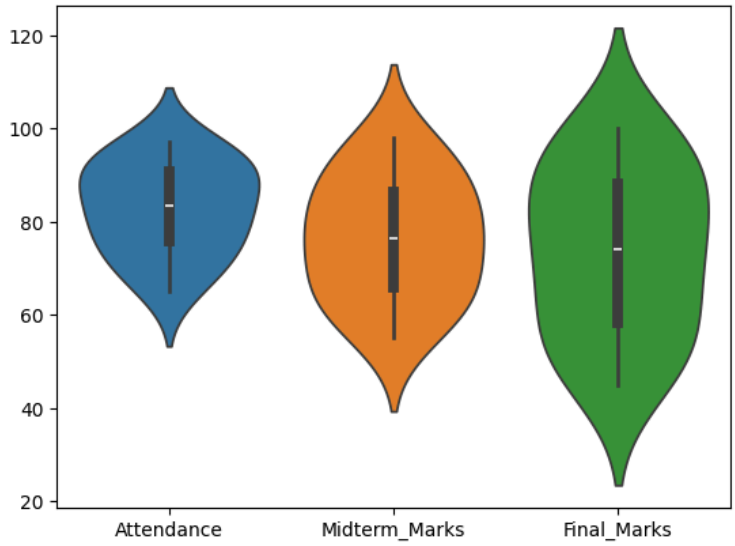


Step - 6 create violine plot

```
sns.violinplot(y="Study_Hours", data=df)
plt.show()
```




```
#create violin plots for Attendance Misrerm_Marks and Final_Marks.
sns.violinplot(data=df[["Attendance", "Midterm_Marks", "Final_Marks"]])
plt.show()
```



Step 7 - save the dataset

```
# Save the updated dataset .
df.to_csv('Dataset_analysis.csv', index=False)
print("Dataset saved successfully as 'Dataset_analysi.csv'")
```

```
Dataset saved successfully as 'Dataset_analysi.csv'
```

```
# Save the dataset first to ensure the file exists on disk
df.to_csv('Dataset_analysis.csv', index=False)

from google.colab import files
files.download("Dataset_analysis.csv")
```